

INVOLUTIVE NORM DYNAMICS IN SPLIT CYCLIC GROUP ALGEBRAS: FIXED SEQUENCES, SHARP DEPTH, AND RIGIDITY

ANONYMOUS

ABSTRACT. Let $n \mid (q - 1)$ and let inversion on the cyclic group C_n induce the canonical involution $*$ of the split group algebra $\mathcal{A}_{q,n} = \mathbb{F}_q[C_n]$. We determine the complete finite dynamics of the nonlinear norm map $T(a) = aa^*$. Character inversion partitions the Fourier coordinates into self-inverse blocks $z \mapsto z^2$ and paired blocks $(u, v) \mapsto (uv, uv)$. If $s = \gcd(n, 2)$ and $o = (n + s)/2$, then

$$\#\text{Fix}(T^k) = (1 + \gcd(2^k - 1, q - 1))^o$$

for every $k \geq 1$. Writing $q - 1 = 2^\alpha m$ with m odd, the recurrent core has size $(m + 1)^o$, while the maximum transient depth is exactly $\alpha + \mathbf{1}_{\{n > s\}}$. Möbius inversion gives every cycle and a finite product for the Artin–Mazur zeta function. Finally, phase-space size, the fixed sequence, and maximum depth recover (q, n) , including the ambiguous two-orbit branch. Literal cyclic convolution over prime fields and explicit models of $\mathbb{F}_4$ and $\mathbb{F}_{16}$ independently check the formulas.

1. INTRODUCTION AND OWNERSHIP BOUNDARY

Let $q = p^h$ be a prime power, let $n \geq 1$ divide $q - 1$, and choose a generator g of C_n . On

$$\mathcal{A}_{q,n} = \mathbb{F}_q[C_n] = \left\{ \sum_{r=0}^{n-1} a_r g^r : a_r \in \mathbb{F}_q \right\}$$

consider the canonical inversion involution

$$\left(\sum_{r=0}^{n-1} a_r g^r \right)^* = \sum_{r=0}^{n-1} a_r g^{-r}$$

and the self-map

$$(1) \quad T: \mathcal{A}_{q,n} \longrightarrow \mathcal{A}_{q,n}, \quad T(a) = aa^*.$$

The phase space has q^n elements. The assumption $n \mid (q - 1)$ is kept throughout: it makes the cyclic algebra split over the stated ground field and excludes modular nilpotents from $x^n - 1$.

Group-algebra involutions and their symmetric and unitary units are classical; see, for example, Bovdi and Grishkov [2]. Finite Fourier decomposition is standard [4], and scalar power-map functional graphs over finite fields have an established literature [3, 5]. We assign those ingredients no novelty credit. Our bounded target is their conjunction for the map (1) on the *whole* group algebra, including zero divisors: the paired-coordinate synchronization, all fixed and cycle counts, the sharp transient depth, and a recovery statement. A targeted source search completed on 29 August 2026 did not locate this exact package, but that database- and query-bounded observation is not a priority or novelty claim. External circulation remains on **HOLD** pending a specialist review.

There are two proof routes. A coefficient-free involution calculation first collapses every orbit to ordinary powering inside the symmetric algebra. Fourier idempotents then expose independent inversion-orbit blocks and make the census exact. The computational control reverses this order: it starts from literal coefficient convolution and cyclic reversal, then compares the result with the Fourier and arithmetic predictions.

Date: Internal Stage 2 draft, 29 August 2026.

2. ONE-STEP COLLAPSE AND FOURIER BLOCKS

Write

$$\mathcal{A}_{q,n}^+ = \{b \in \mathcal{A}_{q,n} : b^* = b\}.$$

The first route requires neither a primitive root nor a choice of coordinates.

Proposition 2.1 (Algebraic collapse). *For every $a \in \mathcal{A}_{q,n}$,*

$$T(a) \in \mathcal{A}_{q,n}^+, \quad T^k(a) = (aa^*)^{2^{k-1}} \quad (k \geq 1).$$

Moreover $T(b) = b^2$ for every $b \in \mathcal{A}_{q,n}^+$.

Proof. The cyclic group algebra is commutative, so $(aa^*)^* = a^*a = aa^*$. Thus $T(a)$ is symmetric. If $b = b^*$, then $T(b) = bb^* = b^2$. Starting with $b = aa^*$ and iterating squaring proves the displayed formula. $\square$

For the second route, choose a primitive n th root $\omega \in \mathbb{F}_q$. The Fourier transform

$$\widehat{a}(j) = \sum_{r=0}^{n-1} a_r \omega^{jr}, \quad j \in \mathbb{Z}/n\mathbb{Z},$$

is an algebra isomorphism from $\mathcal{A}_{q,n}$ to $\mathbb{F}_q^n$. Indeed, $p \nmid n$ and the usual Fourier matrix is invertible. Multiplication becomes coordinatewise, while

$$\widehat{a^*}(j) = \widehat{a}(-j).$$

Let

$$(2) \quad s = \gcd(n, 2), \quad o = \frac{n+s}{2}.$$

Here s is the number of self-inverse characters and o is the number of orbits of $j \mapsto -j$.

Theorem 2.2 (Fourier block normal form). *Under the Fourier isomorphism, a self-inverse character carries the block*

$$z \mapsto z^2,$$

whereas a two-element orbit $\{j, -j\}$ carries

$$(u, v) \mapsto (uv, uv).$$

In particular, for every $k \geq 1$ and every j ,

$$(3) \quad \widehat{T^k(a)}(j) = (\widehat{a}(j)\widehat{a}(-j))^{2^{k-1}}.$$

The fixed algebra $\mathcal{A}_{q,n}^+$ is isomorphic, as an algebra, to $\mathbb{F}_q^o$.

Proof. Coordinatewise multiplication gives

$$\widehat{T(a)}(j) = \widehat{a}(j)\widehat{a}(-j).$$

If $j = -j$, this is z^2 . Otherwise the two output coordinates coincide, giving the paired block. After the first step both kinds of blocks square, which proves (3). Finally, $a = a^*$ is equivalent to $\widehat{a}(j) = \widehat{a}(-j)$ on every inversion orbit; choosing one common coordinate on each orbit identifies the fixed algebra with $\mathbb{F}_q^o$. $\square$

The paired block exhibits the first qualitative anomaly: two independent input coordinates synchronize in one step and never separate. This is the only source of an extra transient level beyond scalar squaring.

3. FIXED POINTS AND THE COMPLETE RECURRENT CENSUS

For $k \geq 1$, put

$$(4) \quad L_k = 1 + \gcd(2^k - 1, q - 1).$$

Theorem 3.1 (Exact fixed sequence). *For every $k \geq 1$,*

$$(5) \quad F_k := \# \text{Fix}(T^k) = L_k^o = (1 + \gcd(2^k - 1, q - 1))^o.$$

Equivalently, $T^k(a) = a$ precisely when $a \in \mathcal{A}_{q,n}^+$ and each of its o symmetric Fourier coordinates satisfies $z^{2^k} = z$.

Proof. In a paired block, fixedness under the k th iterate and [equation \(3\)](#) force

$$u = v = (uv)^{2^{k-1}}.$$

Writing the common value as z reduces the condition to $z^{2^k} = z$. The same equation holds on a self-inverse block. It has the root 0 and exactly $\gcd(2^k - 1, q - 1)$ roots in the cyclic group $\mathbb{F}_q^\times$. The o inversion orbits are independent, proving [\(5\)](#). The equivalent description also follows directly from [proposition 2.1](#): a fixed point of T^k lies in the first image and hence is symmetric, where T^k is 2^k th powering. $\square$

Write

$$(6) \quad q - 1 = 2^\alpha m, \quad m \text{ odd},$$

and let $\mu_m \leq \mathbb{F}_q^\times$ be the subgroup of order m . Set

$$R_q = \{0\} \cup \mu_m.$$

For a point x in a finite dynamical system, define $\text{depth}(x) = \min\{r \geq 0 : T^r(x) \text{ is periodic}\}$, and let D be the maximum of this depth over the phase space.

Theorem 3.2 (Recurrent core and sharp depth). *The recurrent set is, in symmetric Fourier coordinates,*

$$(7) \quad \text{Rec}(T) = R_q^o \subseteq \mathcal{A}_{q,n}^+.$$

Consequently

$$(8) \quad \# \text{Rec}(T) = (m + 1)^o.$$

The maximum transient depth is

$$(9) \quad D = \alpha + \mathbf{1}_{\{n > s\}}.$$

Both bounds are attained, including the cases $n = 1, 2$ and characteristic two.

Proof. For $z \neq 0$, write $\text{ord}(z) = 2^u d$ with d odd. Repeated squaring removes one factor of 2 from the order at each step until $u = 0$. Thus z is periodic under squaring if and only if $z \in \mu_m$; zero is fixed. Its scalar depth is u , whose maximum is α , attained by a primitive element of $\mathbb{F}_q^\times$.

A periodic paired block must already be in the image of its update, hence must be diagonal. On the diagonal it is scalar squaring, so its recurrent points are exactly (z, z) with $z \in R_q$. This proves [\(7\)](#) and [\(8\)](#).

Every self-inverse block reaches its recurrent part in at most α steps. Every paired block becomes diagonal after one step and then needs at most α further squarings. Hence the right side of [\(9\)](#) is an upper bound. More precisely, if $d_q(z)$ denotes scalar depth under $z \mapsto z^2$, then a self block z has depth $d_q(z)$, while a paired block (u, v) has depth

$$\text{depth}(u, v) = \begin{cases} d_q(u), & u = v, \\ 1 + d_q(uv), & u \neq v. \end{cases}$$

Indeed, diagonal pairs already follow scalar squaring, whereas a nondiagonal pair cannot be periodic and first synchronizes to (uv, uv) . Thus the extra level belongs only to genuinely

TABLE 1. Exact fixed-point signals; entries list $F_1, F_2, \dots$

q	n	o	first fixed counts
7	3	2	4, 16, 4, 16, 4, 16
11	2	2	4, 4, 4, 36, 4, 4
5	4	3	8, 8, 8, 8, 8, 8
16	3	2	4, 16, 4, 256, 4, 16

nondiagonal pairs. If no paired orbit exists, then $n = s$, equivalently $n \in \{1, 2\}$, and a primitive scalar coordinate attains depth α . If a pair exists, put $(1, \gamma)$ on that block for a primitive $\gamma \in \mathbb{F}_q^\times$ and put recurrent values on the other blocks. The first step gives (γ, γ) , which still has scalar depth α ; the total depth is $\alpha + 1$. When q has characteristic two, $\alpha = 0$; a pair can occur only when $n \geq 3$, and the same construction attains the remaining one synchronization step. $\square$

4. CYCLES AND THE ARTIN–MAZUR ZETA FUNCTION

Let P_k be the number of points of least period k and $C_k = P_k/k$ the number of k -cycles. Let μ denote the Möbius function.

Theorem 4.1 (All cycles and zeta). *For every $k \geq 1$,*

$$(10) \quad P_k = \sum_{d|k} \mu(k/d) (1 + \gcd(2^d - 1, q - 1))^o,$$

$$(11) \quad C_k = \frac{1}{k} \sum_{d|k} \mu(k/d) (1 + \gcd(2^d - 1, q - 1))^o.$$

If $\ell = \text{ord}_m(2)$ for $m > 1$ and $\ell = 1$ for $m = 1$, then $C_k = 0$ unless $k \mid \ell$. The Artin–Mazur zeta function [1] is the rational finite product

$$(12) \quad \zeta_T(z) := \exp \left(\sum_{k \geq 1} \frac{F_k}{k} z^k \right) = \prod_{k|\ell} (1 - z^k)^{-C_k}.$$

In particular, (11) is a complete cycle inventory, not only an asymptotic count.

Proof. In any finite map, $F_k = \sum_{d|k} P_d$. Möbius inversion and [theorem 3.1](#) give (10); division by k gives (11). On $R_q \setminus \{0\} = \mu_m$, squaring is an automorphism of order ℓ . Coordinatewise squaring on R_q^o also has order ℓ , so every recurrent period divides ℓ . Finally, a cycle of length k contributes

$$\exp \left(\sum_{r \geq 1} \frac{z^{rk}}{r} \right) = (1 - z^k)^{-1}$$

to the defining exponential. Multiplying over all cycles proves (12). $\square$

The small fixed sequences in [table 1](#) show that the arithmetic is visible immediately. The contrast between $q = 7$ and $q = 16$ at time four, for example, detects different odd parts of $q - 1$ although both begin 4, 16, 4 when $n = 3$.

5. RIGIDITY FROM TEMPORAL DATA

The exponent o remembers only the number of inversion orbits, so by itself it leaves an odd/even ambiguity in n . Depth and phase size resolve it.

Theorem 5.1 (Recovery of the field size and cycle length). *Within the family $n \mid (q - 1)$, the data*

$$(13) \quad (q^n, (F_k)_{k \geq 1}, D)$$

determine the ordered pair (q, n) uniquely. More explicitly:

- (1) $o = \log_2 F_1$;
- (2) $m = \max_{k \geq 1} F_k^{1/o} - 1$;
- (3) the only possible lengths are $n = 2o - 1$ and, when $o \geq 2$, $n = 2o - 2$;
- (4) for a candidate n , put

$$\varepsilon_n = \mathbf{1}_{\{n > \gcd(n, 2)\}}, \quad q_n = 2^{D - \varepsilon_n} m + 1.$$

The unique candidate satisfying the phase-size identity $q_n^n = q^n$ (and the family conditions) is the original pair.

Proof. Since $L_1 = 2$, [theorem 3.1](#) gives $F_1 = 2^o$. Because $2^k - 1$ is odd,

$$\gcd(2^k - 1, q - 1) = \gcd(2^k - 1, m).$$

If $m > 1$, some k satisfies $2^k \equiv 1 \pmod{m}$; for $m = 1$ the claim is immediate. Thus the maximum in item 2 is exactly m . Solving $2o = n + \gcd(n, 2)$ gives precisely the candidates in item 3, and [theorem 3.2](#) gives the formula for q_n .

It remains to show that the candidates cannot both survive. If $o = 1$, only $n = 1$ is positive. If $o \geq 3$, both candidate lengths have paired characters, so both have $\varepsilon_n = 1$ and hence the same candidate field size. Their distinct positive exponents cannot give the same phase size.

The only delicate branch is $o = 2$, where the candidates are $n = 2$ and $n = 3$. Write their reconstructed field sizes as q_2 and q_3 . Their different synchronization corrections give

$$q_2 - 1 = 2(q_3 - 1).$$

If both also matched the phase size, then $q_2^2 = q_3^3$, so coprimality of 2 and 3 would give integers $q_2 = r^3$ and $q_3 = r^2$ with $r \geq 2$. Substitution yields

$$r^3 - 1 = 2(r^2 - 1), \quad (r - 1)(r^2 - r - 1) = 0,$$

which has no integer solution with $r \geq 2$. Hence at most one candidate survives in every branch, while the true one necessarily does. $\square$

Remark 5.2 (Scope of rigidity). *The phase-size datum in (13) is the cardinality of the given finite phase space, not an invariant inferred from the zeta function. The theorem is therefore a qualified recovery statement inside the split cyclic family, not a claim that zeta functions classify arbitrary finite maps or group algebras.*

6. EXACT CONTROLS AND LIMITATIONS

The accompanying verifier uses no computer-algebra package. For each stored small lane it enumerates all coefficient vectors, computes cyclic convolution and reversal literally, constructs the full functional graph, and checks: the Fourier product rule, [equations \(3\), \(5\), \(8\) and \(9\)](#), the Möbius cycle ledger, and the zeta exponents. Prime-field lanes include $(q, n) = (3, 1), (3, 2), (5, 2), (5, 4), (7, 3), (11, 2), (13, 3)$. Separate polynomial basis implementations of

$$\mathbb{F}_4 = \mathbb{F}_2[t]/(t^2 + t + 1), \quad \mathbb{F}_{16} = \mathbb{F}_2[t]/(t^4 + t + 1)$$

exercise genuine extension-field arithmetic at $n = 3$. A broader arithmetic lane checks the recovery algorithm for all divisors $n \mid (q - 1)$ over a registered list of prime powers. The stored output is a deterministic exact certificate, not numerical evidence in place of the proofs.

The splitting hypothesis is essential to the present formulas. When $n \nmid (q - 1)$, Frobenius acts nontrivially on character blocks; when $p \mid n$, the group algebra is not semisimple. Neither regime is covered here. Likewise, the cited unit literature focuses largely on units, whereas the transient maximum here is often attained using a whole-algebra block; this distinction motivates the calculation but does not establish priority.

There is also an internal firewall against P86. That paper studies the spatial two-block factor $Y_i = U_i U_{i+1}$ of an iid bi-infinite field process, with support entropy and prediction memory as its observables. Here a paired product is only one Fourier block of a finite self-map; after one update it synchronizes and enters scalar squaring, and the observables are temporal

fixed counts, cycles, and depth. The shared multiplication primitive is therefore disclosed rather than counted as a separate contribution.

DATA AND CODE AVAILABILITY

The exact verifier, its stored output, claim-to-evidence ledger, and build instructions accompany this internal manuscript. No external data are used.

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